—— AZ-104 // NETWORKING

Implement & Manage VM Networks

NICs, public IPs, DNS at the VM level, IP forwarding, Accelerated Networking, Network Watcher diagnostics, and how all of a VM's network configuration fits together.

What makes up a VM's network configuration

Every VM connects to a VNet through a Network Interface Card (NIC). The NIC holds the private IP, optional public IP, NSG association, and DNS settings. Understanding what lives on the NIC vs the VM vs the subnet is the key to getting VM networking questions right on the exam.

Network Interface Card (NIC)

nic-vm-web01 — every VM has at least one

PRIVATE IP 10.0.1.4 Static or Dynamic — assigned from subnet range	PUBLIC IP 20.10.50.200 Optional — Basic or Standard SKU
SUBNET web-subnet Cannot change after VM creation without deallocation	NSG nsg-web (optional) NIC-level NSG evaluated after subnet NSG (inbound)
DNS SERVERS Custom or Azure default Overrides VNet-level DNS for this NIC only	IP FORWARDING Disabled (default) Enable for NVAs that route traffic for others

Eight terms you need cold

NIC (Network Interface Card)

The Azure resource that connects a VM to a VNet subnet. A VM can have multiple NICs (count depends on VM size). Each NIC can have multiple IP configurations (primary + secondary private IPs).

Tip: You can add or remove secondary NICs on a deallocated VM — but you cannot change the primary NIC's subnet without rebuilding.

Public IP Address

An Azure resource (not a property of the NIC) assigned to a NIC or load balancer frontend. Two SKUs: Basic (dynamic, no zone support) and Standard (static, zone-redundant).

Tip: Standard SKU is required for zone-redundant load balancers. Basic Public IPs are being retired — use Standard for all new deployments.

Static vs Dynamic IP

Dynamic private IPs are assigned by Azure's DHCP and may change on deallocation. Static private IPs persist across deallocations — required for DNS servers, domain controllers, and firewalls.

Tip: Set the IP to Static *after* Azure assigns it dynamically — pick the already-assigned IP to avoid conflicts.

IP Forwarding

Allows a NIC to accept traffic addressed to IPs it doesn't own — required for NVAs (Network Virtual Appliances) and VMs acting as routers or firewalls.

Tip: Enable IP forwarding on the NIC in Azure AND configure the OS to forward IP packets — both are required.

Accelerated Networking

Bypasses the hypervisor network stack by using SR-IOV (Single Root I/O Virtualization) — dramatically reduces latency and CPU overhead for network-intensive workloads.

Tip: Requires a supported VM size and OS. Enable it at NIC creation — you can't turn it on for an existing NIC without recreation.

Network Watcher

A suite of diagnostic tools for Azure networking: IP flow verify, next hop, connection troubleshoot, packet capture, NSG flow logs, topology viewer.

Tip: Must be enabled per region — it's auto-enabled when you create a VNet but worth confirming before trying to use its tools.

IP Flow Verify

A Network Watcher tool that tests whether a specific packet (source IP, dest IP, port, protocol) is allowed or denied by NSG rules — tells you exactly which rule is responsible.

Tip: The fastest way to diagnose "why is port X blocked" without reading through every NSG rule manually.

Next Hop

A Network Watcher tool that shows the next hop for traffic from a VM to a destination IP — reveals whether traffic goes through Azure default routing, a UDR, or an NVA.

Tip: Use Next Hop to verify that UDRs are working as expected — if traffic should go to a firewall but Next Hop shows Internet, the route table isn't applied correctly.

PUBLIC IP SKU	ALLOCATION	ZONE SUPPORT	IDLE TIMEOUT	RETIRE DATE
Basic	Dynamic or Static	No	4–30 min	Sept 2025 — avoid for new deployments
Standard	Static only	**Zone-redundant**	4 min (fixed)	**Recommended for all new deployments**

EXAM GOTCHA — BASIC PUBLIC IP RETIREMENT

Microsoft announced the retirement of Basic SKU Public IP addresses. The exam still tests both SKUs, but **Standard is always the correct choice for new deployments**. Basic does not support zone redundancy, cannot be used with Standard SKU Load Balancers, and has an open default inbound access policy (Standard blocks all inbound by default — more secure).

Configuring and troubleshooting VM network connectivity

Use this flow when setting up a new VM's networking or diagnosing why a VM can't reach something.

START

You need to configure or troubleshoot a VM's network connection

DECIDE

New VM setup or troubleshooting an existing one?

NEW SETUP

Continue to NIC configuration decisions below

TROUBLESHOOTING

Jump to Network Watcher — IP Flow Verify first, then Next Hop

DECIDE

Does this VM need a public IP?

YES

Create Standard SKU Public IP (Static) — assign to NIC

NO

Private IP only — use Bastion or VPN for management access

DECIDE

Does the private IP need to be stable (DNS server, DC, firewall)?

YES

Set private IP allocation to Static — pick the already-assigned address

NO

Dynamic is fine

DECIDE

Does this VM route traffic for other resources (NVA / router)?

YES

Enable IP Forwarding on NIC in Azure AND configure OS IP forwarding

NO

Leave IP Forwarding disabled

DECIDE

Is network performance critical (low latency, high throughput)?

YES

Enable Accelerated Networking on the NIC (requires supported VM size + OS)

NO

Standard networking is sufficient

VERIFY

Use Network Watcher IP Flow Verify to confirm NSG rules allow expected traffic

DONE

VM network configured, IPs set, forwarding and Accelerated Networking enabled as needed

Tools for VM network management

>_ Azure CLI — NIC & Public IP

BEST FOR: scripted NIC creation and IP config

Create and configure NICs, attach public IPs, and update IP allocation methods in pipeline scripts.

```
az network public-ip create \
  -g rg-prod -n pip-vm-web01 \
  --sku Standard --allocation-
method Static

az network nic create \
  -g rg-prod -n nic-vm-web01 \
  --vnet-name vnet-prod \
  --subnet web-subnet \
  --public-ip-address pip-vm-web01

# Set private IP to static
az network nic ip-config update \
  -g rg-prod --nic-name nic-vm-
web01 \
  -n ipconfig1 \
  --private-ip-address 10.0.1.4 \
  --private-ip-address-version IPv4
```

>_ Azure CLI — IP Forwarding & Accel Networking

BEST FOR: NVA setup, performance optimization

Enable IP forwarding for NVA VMs and turn on Accelerated Networking for performance-sensitive workloads.

```
# Enable IP forwarding
az network nic update \
  -g rg-prod -n nic-nva01 \
  --ip-forwarding true

# Enable Accelerated Networking
az network nic update \
  -g rg-prod -n nic-vm-web01 \
  --accelerated-networking true

# Check effective routes
az network nic show-effective-
route-table \
  -g rg-prod -n nic-vm-web01 \
  --output table
```

Network Watcher — Diagnostics

BEST FOR: troubleshooting connectivity and NSG issues

IP Flow Verify tells you exactly which NSG rule is blocking or allowing traffic. Next Hop tells you where traffic is actually being routed. Connection troubleshoot tests end-to-end connectivity.

```
# IP Flow Verify — is port 443 allowed?
az network watcher test-ip-flow \
  -g rg-prod \
  --vm vm-web01 \
  --direction Inbound \
  --protocol TCP \
  --local 10.0.1.4:443 \
  --remote 203.0.113.0:12345

# Next Hop — where does traffic go?
az network watcher show-next-hop \
  -g rg-prod \
  --vm vm-web01 \
  --source-ip 10.0.1.4 \
  --dest-ip 10.0.3.10
```

Azure CLI — NSG Flow Logs

BEST FOR: auditing and traffic analysis

NSG flow logs capture all traffic allowed and denied by an NSG — essential for security auditing, cost analysis, and diagnosing intermittent connectivity issues.

```
az network watcher flow-log create \ -g rg-prod \ --nsg nsg-web \ --storage-account mystorageacct \ --enabled true \ --retention 30 az network watcher flow-log list \ -g rg-prod --location eastus
```

〈〉 Bicep — NIC + Public IP

BEST FOR: IaC — VM network config as code

Deploy NIC and Public IP together with the VM — everything defined in one template, nothing to configure manually after deployment.

```
resource pip
'Microsoft.Network/publicIPAddresse
s@2023-04-01' = {
  name: 'pip-vm-web01'
  location: location
  sku: { name: 'Standard' }
  properties: {
publicIPAllocationMethod: 'Static'
}
}

resource nic
'Microsoft.Network/networkInterface
s@2023-04-01' = {
  name: 'nic-vm-web01'
  location: location
  properties: {
    enableIPForwarding: false
    enableAcceleratedNetworking:
true
    ipConfigurations: [{
      name: 'ipconfig1'
      properties: {
        subnet: { id: subnetId }
        publicIPAddress: { id:
pip.id }
        privateIPAllocationMethod:
'Static'
        privateIPAddress:
'10.0.1.4'
      }
    }]
  }
}
```

NETWORK WATCHER TOOL	ANSWERS	WHEN TO USE
IP Flow Verify	Is this traffic allowed or denied — and which NSG rule decides?	First step for any "VM can't connect" issue
Next Hop	Where does traffic actually go from this VM?	Verify UDRs are routing through the right NVA/firewall
Connection Troubleshoot	Can VM A reach VM B on port X right now?	End-to-end connectivity test including NSG + routing
Packet Capture	What packets are actually leaving/arriving at this NIC?	Deep inspection — last resort before OS-level tools
NSG Flow Logs	What traffic has been allowed/denied over time?	Auditing, forensics, intermittent issue diagnosis

Principles worth internalizing

EFFECTIVE SECURITY RULES — THE FULL PICTURE

A VM's effective NSG rules are the combination of rules from the **subnet NSG** and the **NIC NSG**. Use the "Effective security rules" view in the portal (VM → Networking → NIC → Effective security rules) to see all rules that apply, merged in priority order — far faster than manually cross-referencing two NSGs during troubleshooting.

EXAM GOTCHA — CHANGING SUBNET AFTER VM CREATION

You **cannot move a VM's NIC to a different subnet** while the VM is running or even stopped (not deallocated). The VM must be **deallocated**, then you can detach the NIC, modify it to point to the new subnet, and reattach it. Moving a NIC to a different VNet is not supported at all — you must delete and recreate the VM in the target VNet.

Six mistakes that show up on the exam

Using Basic SKU public IP with a Standard Load Balancer

Basic and Standard SKU resources cannot be mixed in a load balancer frontend — both the public IP and load balancer must be the same SKU (Standard).

Enabling IP forwarding in Azure but forgetting the OS

IP forwarding must be enabled in both the Azure NIC setting AND inside the OS (e.g. `net.ipv4.ip_forward=1` on Linux) — the Azure setting alone is not enough.

Trying to enable Accelerated Networking on an unsupported VM size

Accelerated Networking requires specific VM sizes (D/E/F series v3+) and a supported OS image. Check compatibility before assuming it's available.

Setting a static IP without first noting the dynamic assignment

When converting from dynamic to static, Azure won't warn you if you pick an IP already in use elsewhere. Always set static to the already-assigned dynamic IP to stay safe.

Forgetting Network Watcher must be enabled per region

Network Watcher is auto-enabled when a VNet is created, but if it was disabled or the region has no VNet yet, tools like IP Flow Verify will fail silently.

Trying to change NIC subnet without deallocating the VM

The VM must be fully deallocated (not just stopped) before a NIC's subnet can be changed. A running or stopped-but-allocated VM will return an error.